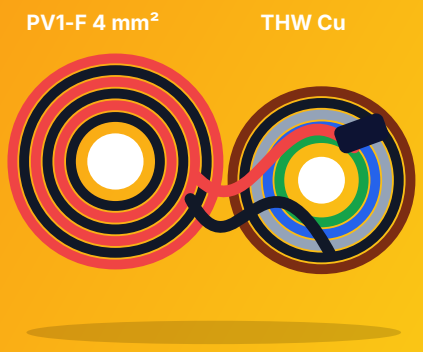
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CABLES · AC THW · DC PV1-F

The right cable on each side

Solar ESS product comparison · October 2026 · Thai Yazaki, BCC, LINK, Lapp, Phelps Dodge



AC SIDE

Thai Yazaki THW copper

TIS 11 / IEC 60227 copper cable, widely stocked. Use copper THW, never aluminium THW-A, on inverter and battery circuits.

DC SIDE

LINK PV1-F 4 mm²

Tinned copper, double insulated, UV-resistant. Choose Lapp or Phelps Dodge H1Z2Z2-K where documented EN 50618 is required.

AC cable vs DC solar cable

Max conductor temperature

THW (AC)	70 °C
PV1-F / H1Z2Z2-K (DC)	**90 °C**

Rated voltage

THW (AC)	750 V
PV1-F (DC)	1000 V
H1Z2Z2-K (DC)	**1500 V**

Full comparison

Cable	Use	Standard	Notes
Thai Yazaki THW 10 / 16 / 35 mm²	AC	**TIS 11 / IEC 60227**	**Copper, widely available**
BCC THW	AC	TIS certified	Copper, retail stock
LINK PV1-F 4 mm²	DC	Tinned copper, XLPE, 1000/1500 V	**Best value and stock**
Lapp Solar 4 mm²	DC	**EN 50618 / IEC 62930**	Documented compliance
Phelps Dodge H1Z2Z2-K	DC	**EN 50618**	Documented compliance

Colour code

DC POSITIVE
Red
PV string +

DC NEGATIVE
Black
PV string –

EARTH
Green / yellow
Frames, rails, boxes

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AC PHASES (THAILAND)
Brown · Black · Grey
L1 · L2 · L3

NEUTRAL
Blue
N

RULE
Never mix
AC and DC in one conduit

Why solar cable on the roof? Ordinary building wire is single-insulated and not UV-rated. PV cable is double insulated, rated 90–120 °C and built for 25 years of sun and rain.

Specifications are from the manufacturers' published datasheets; star ratings are Solar ESS's own assessment. Check the current datasheet before ordering. For prices and a design for your site, book a consultation at www.solaress.org or LINE: solaress1.

Sources: Phelps Dodge H1Z2Z2-K · Lapp